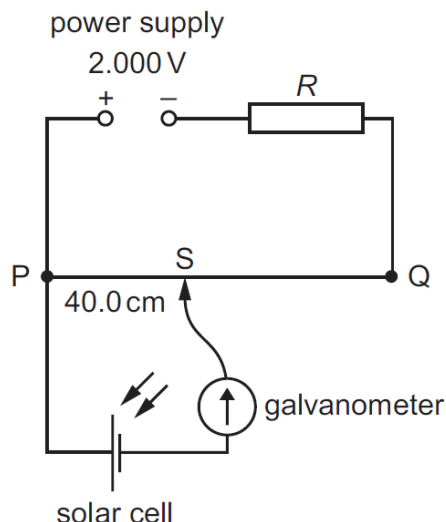


22 A power supply and a solar cell are compared using the potentiometer circuit shown.



The potentiometer wire PQ is 100.0 cm long and has a resistance of 5.00 Ω . The power supply has an e.m.f. of 2.000 V and the solar cell has an e.m.f. of 5.00 mV.

When the galvanometer shows zero deflection, the balance length PS is found to be 40.0 cm.

What is the resistance of R so that the galvanometer reads zero at this balance length?

- | | | | | | | | |
|----------|--------------|----------|--------------|----------|--------------|----------|---------------|
| A | 395 Ω | B | 795 Ω | C | 995 Ω | D | 1055 Ω |
|----------|--------------|----------|--------------|----------|--------------|----------|---------------|

Answer: B

At balance point, voltage across 40.0 cm $\rightarrow 5.00 \times 10^{-3}$ V

Pd across PQ (100 cm) = $\frac{5.00 \times 10^{-3}}{40} \times 100 = 0.0125$

Total voltage across R = 2.000 – 0.0125 V = 1.9875 V

Total resistance = R + 5.00

By potential divider principle: $1.9875 = 2 \frac{R}{R+5}$

$0.99375(R+5) = R$

$0.99375R + 4.96875 = R$

$0.00625 R = 4.96875$

$R = 795 \Omega$

22 A beam of protons enters a velocity selector as shown. The electric field E acts vertically